

Assignment 3

Biometric Attendance System (Face & Fingerprint) – Shruti Building

1. Introduction

A Biometric Attendance System is an advanced IoT-based solution that records attendance using unique human traits such as facial features and fingerprints. The system implemented in Shruti Building uses a multi-modal biometric approach, combining both face recognition and fingerprint scanning to improve reliability and accuracy.

This system works through an IoT framework involving edge devices and cloud integration. It captures data through sensors, processes it using algorithms, stores it securely, and allows access through software dashboards.

2. System Architecture and Working Principle

Process Flow:

Data Collection Processing Storage Access

Data Collection:

- Capturing facial images through a camera
- Scanning fingerprints using biometric sensors

Processing:

- Extracting important features from captured data
- Converting raw input into digital templates using algorithms

Storage:

- Saving processed data in local or cloud databases
- Assigning a unique ID to each individual

Access:

- Admin can view attendance through a dashboard
- Reports can be generated instantly

3. Face Recognition: Devices and Components

- High-resolution cameras for better accuracy
- Infrared (IR) sensors for low-light detection and anti-spoofing
- Depth sensors for 3D facial mapping
- Edge processors for local data processing
- RGB cameras for capturing images

4. Fingerprint Recognition: Sensors and Modules

Types of Sensors:

- Optical Sensor: Uses light, cost-effective

- Capacitive Sensor: Uses electrical signals, more secure

Modules:

- Microcontroller/processor
- Memory for storing fingerprint data

5. Multi-Modal Biometric Integration

- User provides face or fingerprint
- Data processed separately
- Decision based on OR or AND logic

Advantages:

- Better accuracy
- Fewer errors
- Strong security

6. Data Flow

1. Data Acquisition
2. Feature Extraction
3. Matching
4. Storage

7. Networking and Cloud

- Wi-Fi/Ethernet connectivity
- Cloud/local servers

Protocols:

- HTTP/HTTPS
- MQTT
- TCP/IP

Functions:

- Data storage
- Attendance tracking
- Dashboard and reports
- Remote access

8. Key Considerations

Accuracy:

- High precision

Latency:

- Fast response

Security:

- Encryption
- Secure storage
- Limited access

Conclusion

This system is a smart IoT-based solution using face and fingerprint recognition. It ensures secure and efficient attendance management with high accuracy and strong security.